



MI-OTA Based First Order Filters for Biomedical Applications

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Abstract

A low-power, electronically tunable voltage-mode first-order universal filter employing a multiple-input operational transconductance amplifiers (MI-OTAs). It does not require component matching, input matching, or signal inversion, simplifying practical implementation. Designed in 65 nm CMOS at 0.5 V, it consumes 10 nW and achieves a 41.7 dB dynamic range. Experimental results using commercially available LM13600N integrated circuits confirm its practical feasibility and versatility for low-power biomedical applications.

Keywords First-order filter · Operational transconductance amplifier · Multiple-input MOS transistor · Bulk-driven MOS transistor

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1 Introduction

First-order filters are essential building blocks in analog signal processing, widely employed in various applications, such as sinusoidal oscillators and higher-order filter design [5, 9]. Characterized by a single time constant, they realize three fundamental transfer functions: low-pass (LPF), high-pass (HPF), and all-pass (APF). These filters are particularly critical in biomedical signal processing. LPFs suppress high-frequency noise and prevent aliasing before analog-to-digital conversion, while HPFs eliminate baseline drift and motion artifacts common in electrocardiogram (ECG) and electroencephalogram (EEG) signals [11, 25]. APFs are equally vital for phase equalization in multichannel systems, allowing for signal alignment without magnitude degradation [21].

Numerous first-order filters employing various active devices—such as operational amplifiers [18], operational transconductance amplifiers (OTAs) [23, 26]—, current conveyors (CCII, DVCC, DDCC) [20–27, 30], current differencing buffered amplifiers (CDBA) [18], and operational transresistance amplifiers (OTRA) [2], have been reported in the literature. Although these topologies provide versatile filtering capabilities, they generally require relatively high supply voltages (often ≥ 1 V) and exhibit significant power dissipation, which restricts their utility in advanced sub-threshold biomedical applications [18–30].

Recently, several universal filters capable of realizing LPF, HPF, and APF responses have been reported [1, 3, 10, 24]. While these designs represent significant advancements, a design gap persists in achieving low-power efficiency within sub-0.5 V environments. Specifically, existing OTA-based and voltage differencing differential difference amplifier (VDDDA)-based universal filters [13, 14] often trade off circuit simplicity for performance, leading to increased complexity in matching requirements and higher power budgets. Moreover, the filter topologies in [1–10] cannot simultaneously provide both non-inverting and inverting transfer characteristics for the LPF, HPF, and APF in a single topology.

This paper presents a new voltage-mode first-order filter using multiple-input operational transconductance amplifiers (MI-OTAs). Our design introduces a novel approach by employing the multiple-input bulk-driven MOS (MIBD-MOST) technique [15, 17]. This technique is uniquely suited for sub-threshold operation, enabling the realization of nano-Siemens (nS) transconductance values, which are essential for low-cutoff frequency applications in biomedical fields. A similar OTA circuit was previously designed and fabricated using 0.18 μm technology [15]. However, in the design presented here, different transistor flavors available in the 65-nm technology were used in the SC composite transistors to further increase the OTA's voltage gain [28]. The proposed MI-OTA is utilized to implement a filter capable of producing LPF, HPF, and APF transfer functions within a single, compact topology. Unlike conventional designs, this structure eliminates the need for component matching, input matching, or signal inversion, significantly simplifying practical implementation. Moreover, the pole for all responses is electronically tunable, offering high adaptability. Although the filter is based on the MI-OTA reported in [13], it realizes different filtering functions. The design is validated through 65 nm CMOS simulations at a 0.5 V supply

voltage with 10 nW power consumption, and its practical feasibility is demonstrated via a prototype using commercial LM13600N integrated circuits.

2 Proposed Circuit

2.1 MI-OTA

The electrical symbol of the MI-OTA is shown in Fig. 1. Ideally, its I-V characteristic is described by the following equation:

$$I_o = g_m(V_{1+} - V_{1-} + V_{2+} - V_{2-} + V_{3+} - V_{3-}) \quad (1)$$

where I_o is the output current, g_m is the circuit transconductance, and V_+ and V_- represent the non-inverting and inverting input terminals, respectively. As seen from Eq. (1), the output current I_o is proportional to the difference between the sum of the voltages applied to the non-inverting and inverting inputs.

The schematic of the CMOS OTA designed and optimized for the application considered in this work is shown in Fig. 2. The circuit can be regarded as a current-mirror OTA whose input stage is linearized using the Krummenacher–Joehl technique [17], adapted for bulk-driven transistors operating in the weak-inversion region. The figure illustrates the implementation of the resulting 2-port MI-OTA, which can be readily extended to a 3-port configuration (e.g., V_{3+} and V_{3-}) or to higher-order input structures by incorporating additional C_B – M_R shunt networks at the input, without increasing the number of MOS transistors in the core OTA.

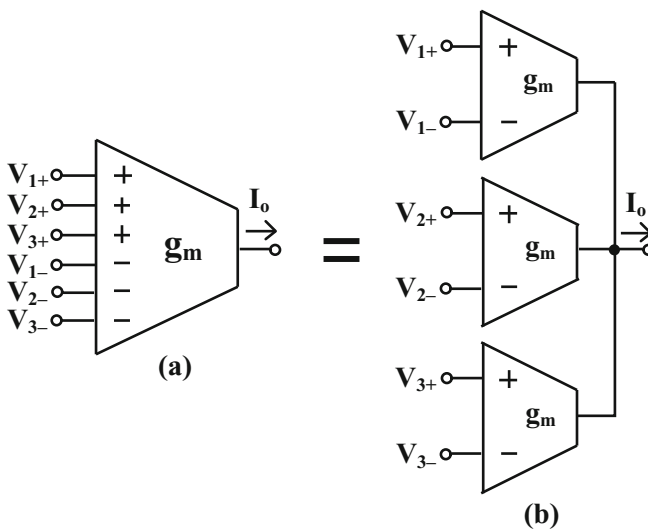


Fig. 1 MI-OTA: **a** electrical symbol, **b** implementation using conventional OTAs

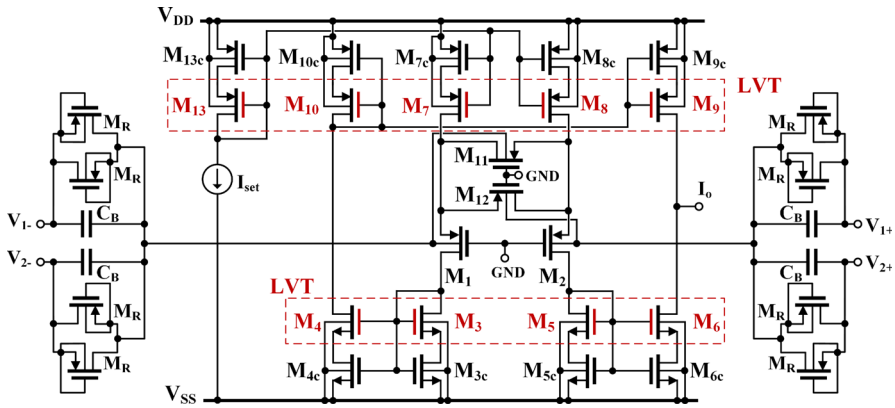


Fig. 2 structure of CMOS 2-port MI-OTA

Transistors M_{11} and M_{12} operate in a triode region and, together with M_1 - M_2 form a linearized input differential stage, biased by the current sources M_{7c} , M_{8c} . Note that all transistors in this design were replaced by self-cascode (SC) composite transistors, which increases the output resistance of the OTA and, consequently its voltage gain. The multiple inputs of the OTA are realized through an input capacitive divider with capacitors C_B with no additional power consumption. The capacitors are shunted by large resistances realized using MOS transistors operating in the cutoff region with $V_{GS} = 0$. These resistances provide proper DC biasing for the input terminals of the OTA.

The cascoding transistors in the SC connections are low-threshold voltage (LVT) transistors, ($V_{TH} = +266/-268$ mV), while the main transistors in SCs are regular ones with V_{TH} of $410/-406$ mV. This configuration ensures that the “bottom” transistors in the self-cascodes operate with a relatively large V_{DS} of approximately 100 mV, at the boundary between the saturation and non-saturation regions, functioning more like standard cascodes than traditional self-cascodes. This design approach allows for maintaining sufficient voltage gain with the 65-nm technology, which typically suffers from lower intrinsic voltage gains in MOS transistors, especially when controlled by the bulk terminal and further degraded by the input capacitive dividers.

The quasi-static I-V large-signal characteristic from one pair of inputs (V_{i+} , V_{i-} , $i = 1, 2$), with the second pair grounded, can be described by the equation [15]:

$$I_{out} = 2I_{set} \tanh\left(\beta\eta \frac{V_{i+} - V_{i-}}{2n_p U_T} - \tanh^{-1}\left[\frac{1}{4m+1} \tanh\left(\beta\eta \frac{V_{i+} - V_{i-}}{2n_p U_T}\right)\right]\right) \quad (2)$$

here n_p is the subthreshold slope factor for a p-channel transistor, U_T is the thermal potential, $\eta = (n_p - 1) = g_{mb(M1,2)}/g_{m(M1,2)}$, $m = (W/L)_{M11,12}/(W/L)_{M1,2}$ is the relative aspect ratio of the two matched transistor pairs $M_{11,12}$ — $M_{1,2}$ and β is the voltage gain of the input capacitive divider, which for identical capacitors C_B and neglecting second-order effects is equal to 0.5. For optimal linearity the coefficient

m should be equal to 0.5, the same as for a gate-driven version of the input stage in weak-inversion.

From (2), the small-signal transconductance of the OTA is:

$$g_m = \beta \cdot \eta \cdot \frac{4m}{4m + 1} \cdot \frac{I_{set}}{n_p U_T} \quad (3)$$

which for $\beta = 0.5$ and $m = 0.5$ yields:

$$g_m = \eta \cdot \frac{1}{3} \cdot \frac{I_{set}}{n_p U_T} \quad (4)$$

thus, the g_m is proportional to the I_{set} . The low-frequency voltage gain of the OTA is given by:

$$A_{VO} \cong g_m r_{out} \quad (5)$$

where r_{out} is its output resistance given by:

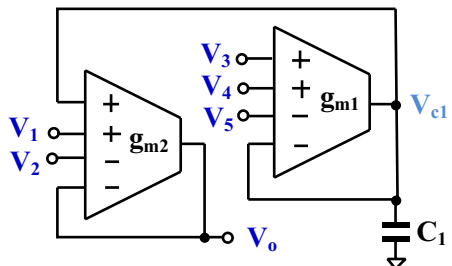
$$r_{out} \cong [(g_{m9} r_{ds9} r_{ds6}) || (g_{m6} r_{ds6} r_{ds9})] \quad (6)$$

The utilization of SC connections significantly enhances the voltage gain, particularly in the architecture where both transistors are biased at the edge of saturation. This configuration effectively mitigates the gain reduction caused by the input capacitive divider and the inherently low transconductance of the BD transistors. From (2) it is worth noting, that applying voltage attenuation at input terminals by the capacitive divider increases linear range of the OTA, but obvious consequence is also increase of its input referred noise, in the same proportion. Consequently, the dynamic range is preserved, matching that of the bulk-driven OTA without the use of capacitive dividers.

2.2 First-Order Filter

To demonstrate the advantages of the multiple-input capability of the OTA, the first-order universal filter is realized, as shown in Fig. 3. The circuit is implemented with two two-input differential OTAs and one grounded capacitor, offering high input

Fig. 3 First-order universal filter employing two 2-port MI-OTAs



impedance and making it highly suitable for voltage-mode applications. With this MI-OTA-based configuration, different filtering responses such as LPF, HPF, and APF can be conveniently obtained. This flexibility is achieved merely by selecting the appropriate input signals, without requiring any changes to the circuit topology. The proposed topology employs an LPF as the fundamental building block. By applying suitable additions and subtractions of the input signals, HPF and APF responses can be obtained from the LPF-based structure. Consequently, the proposed filter realizes multiple filtering functions without requiring conventional low-pass to high-pass or low-pass to all-pass transformation networks.

By applying nodal analysis and using (1), the node voltages V_{c1} and V_o can be expressed as $V_{c1} = \frac{g_{m1}}{sC_1 + g_{m1}}(V_3 + V_4 - V_5)$, $V_o = V_1 - V_2 + V_{c1}$, respectively. These expressions are obtained under the assumption that R_{o1} and R_{o2} are very large, while C_{o1} and C_{o2} are negligibly small, where R_{oj} and C_{oj} denote the parasitic parameters of g_{mj} , for $j = 1, 2$. The output voltage V_o of the filter in Fig. 3 can be expressed as

$$V_o = \left\{ \frac{g_{m1}}{sC_1 + g_{m1}}(V_3 + V_4 - V_5) \right\} + (V_1 - V_2) \quad (7)$$

Here V_1 to V_5 represent the input voltages.

According to Eq. (7), various filtering responses can be realized by properly assigning the input signals, as outlined in Table 1 (where unused inputs are grounded). The results clearly demonstrate that the universal filter offers both non-inverting and inverting LPF and HPF responses, whereas only a non-inverting APF response is obtained. By increasing the number of inputs of the MI-OTA, both non-inverting and inverting APF responses can be realized, as shown in Fig. 4. This filter employs a 2-port and 3-port MI-OTA. An attractive feature of the OTA is that increasing the number of differential inputs does not lead to additional power consumption, since only extra $C_B - M_R$ shunt networks are incorporated at the OTA input terminals.

By applying the same procedure as used for Fig. 3 and under the same assumptions, the voltage V_o in Fig. 4 can be expressed as follows:

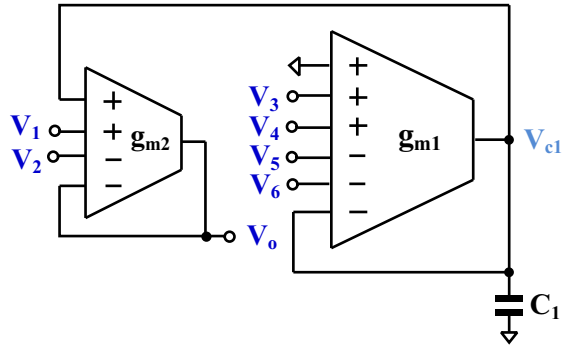
$$V_o = \left\{ \frac{g_{m1}}{sC_1 + g_{m1}}(V_3 + V_4 - V_5 - V_6) \right\} + (V_1 - V_2) \quad (8)$$

Table 1 Input–output configuration for different filtering functions of Figs. 3 and 4

Filtering function		Input	Transfer functions
LPF	Non-inverting	V_3 or $V_4 = V_{in}$	$g_{m1}/D(s)$
	Inverting	V_5 or $V_6 = V_{in}$	$-g_{m1}/D(s)$
HPF	Non-inverting	$V_1 = V_5 = V_{in}$	$sC_1/D(s)$
	Inverting	$V_2 = V_3 = V_{in}$	$-sC_1/D(s)$
APF	Non-inverting	$V_2 = V_3 = V_4 = V_{in}$	$-(sC_1 - g_{m1})/D(s)$
	Inverting	$V_1 = V_5 = V_6 = V_{in}$	$(sC_1 - g_{m1})/D(s)$

$$D(s) = sC_1 + g_{m1}$$

Fig. 4 First-order universal filter employing a 2-port and 3-port MI-OTA



Here V_1 to V_6 represent the input voltages.

As outlined in Table 1, multiple filtering responses can be realized. Accordingly, the configuration shown in Fig. 4 is capable of delivering both non-inverting and inverting transfer functions for the LPF, HPF, and APF by merely extending the number of OTA inputs.

From Eqs. (7) and (8), the pole frequency of the filters is given by:

$$\omega_o = \frac{g_{m1}}{C_1} \quad (9)$$

Thus, the pole frequency can be electronically controlled by g_{m1} . It is evident that the use of a grounded capacitor simplifies the circuit's compensation for parasitic passive capacitances, as such capacitors exhibit lower and more predictable parasitic effects compared to floating configurations due to their connection to a fixed reference potential. The output impedance (z_o) of the circuit is given by $z_o = 1/g_{m2}$.

2.3 Non-ideal Effects

The non-ideal behavior of the OTA is taken into account in the analysis of the first-order universal filter, with particular emphasis on the effects of non-ideal transconductance. Parasitic impedances are neglected, given that the circuit operates at the low frequencies typical of biomedical signals. Under these non-ideal conditions, the output current I_o can be expressed as:

$$I_o = g_{mn}(V_{1+} + V_{2+} - V_{1-} - V_{2-}) \quad (10)$$

Here, g_{mn} represents the non-ideal transconductance of the OTA. At a frequency near its cut-off, g_{mn} can be expressed [22] by:

$$g_{mn}(s) \cong g_m(1 - sT_o) \quad (11)$$

where $T_o = 1/\omega_{gm}$ and ω_{gm} represents the first pole frequency of g_m .

Using Eq. (11), denominator ($D(s)$) of Eq. (8) is given by

$$D(s) = sC_1 \left(1 - \frac{g_{m1}T_o}{C_1} \right) + g_{m1} \quad (12)$$

From (12), the non-ideal effects of the transconductance g_{m1} can be minimized by ensuring that the following conditions are satisfied:

$$\frac{g_{m1}T_o}{C_1} \ll 1 \quad (13)$$

This can be accomplished by properly choosing the values of g_{m1} and the capacitor C_1 .

3 Results

3.1 Simulation Results

The circuit was designed and simulated in the Cadence Virtuoso environment, utilizing the 65 nm TSMC CMOS technology. The layout of the MI-OTA is shown in Fig. 5, occupying a total silicon area of $125 \mu\text{m} \times 92 \mu\text{m}$. Owing to the compact layout

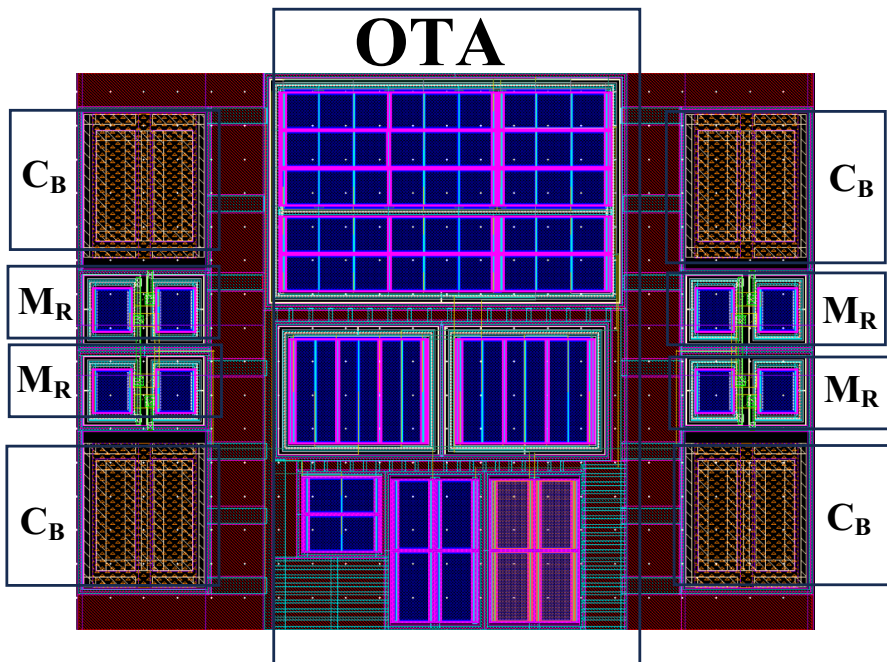


Fig. 5 Layout of the 2-port MI-OTA

Table 2 Component parameters of the MI-OTA

Component	W/L ($\mu\text{m}/\mu\text{m}$)
M_1, M_2	$2 \times 30/3$
M_{11}, M_{12}	30/3
$M_3, M_4, M_5, M_6, M_{3c}, M_{4c}, M_{5c}, M_{6c}$	10/5
$M_7, M_8, M_9, M_{10}, M_{13}$	15/5
$M_{7c}, M_{8c}, M_{9c}, M_{10c}, M_{13c}$	$2 \times 15/5$
M_R	6/5
$C_B = 400 \text{ fF}$	

and low-frequency operation, the influence of parasitic layout effects was minimal. Consequently, the pre-layout and post-layout simulation results showed a strong correlation, confirming the robustness of the design. The component parameters used in the design are summarized in Table 2. The circuit operates with a supply voltage of 0.5 V, ensuring low-power performance suitable for energy-efficient applications. The power consumption of the MI-OTA is simulated at 5 nW, with a bias current setting I_{set} of 2 nA. This low power consumption demonstrates the efficiency of the design, which is essential for battery-powered or energy-sensitive systems.

Figure 6 presents the DC transfer characteristics and transconductance of the MI-OTA as functions of input voltage for different bias currents. The results indicate high linearity and a wide input voltage range, both of which are essential for ensuring stable performance under varying operating conditions.

Figure 7 shows the frequency characteristics of the output current of the MI-OTA for various setting currents. The frequency response of the MI-OTA extends up to approximately 10 kHz, demonstrating its ability to operate effectively within this frequency range.

The filter application from Fig. 3 was designed using a single grounded capacitor of 20 pF, making it suitable for integration into compact systems. Figure 8 illustrates the frequency characteristics of the voltage gain and phase response of the filter for

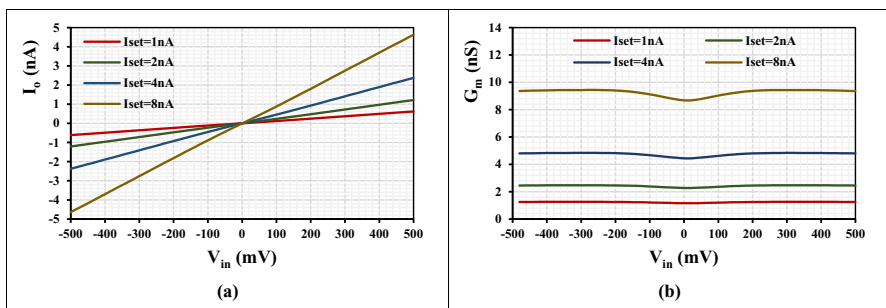


Fig. 6 DC transfer characteristics of the MI-OTA: **a** output current and **b** transconductance versus input voltage

Fig. 7 Frequency characteristics of the output current of the MI-OTA

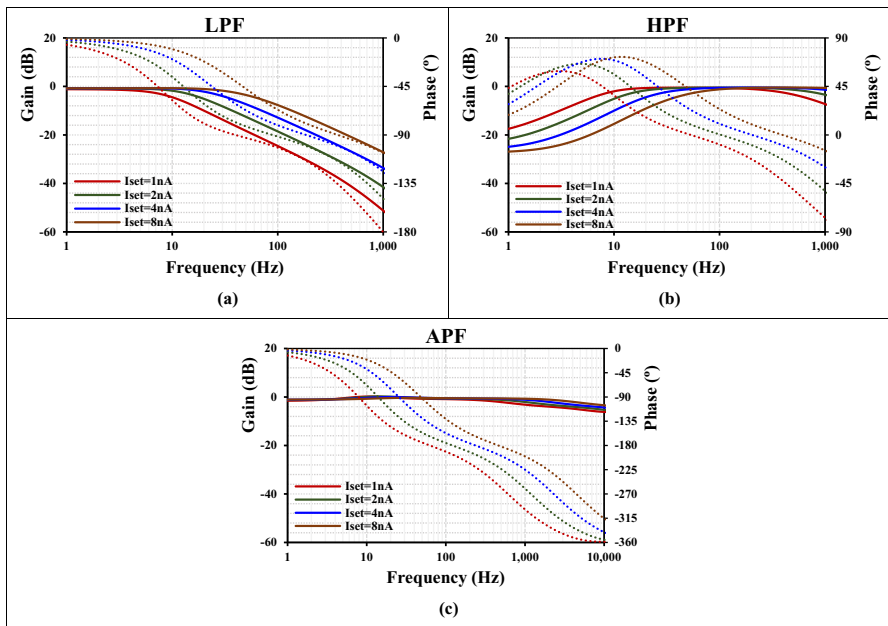
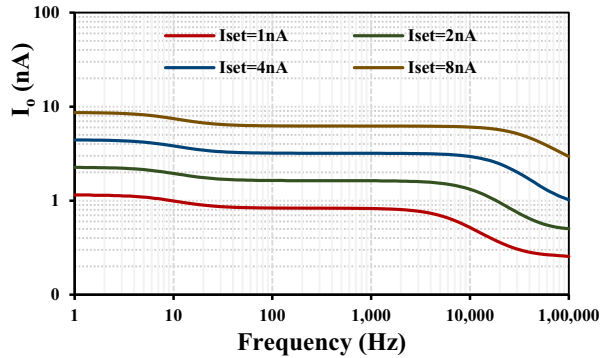


Fig. 8 Frequency characteristics of the voltage gain and phase response of the: **a** LPF, **b** HPF and **c** APF

different types: (a) LPF, (b) HPF, and (c) APF, with various setting currents $I_{set} = (1, 2, 4, 8)$ nA. The corresponding cutoff frequencies are observed at (8 Hz, 14 Hz, 25 Hz, and 48 Hz), respectively, demonstrating the tunability of the filter response based on the setting current. For $I_{set} = 2$ nA, the total power consumption of the filter is 10 nW, confirming its low-power efficiency, which is ideal for energy-constrained applications such as biomedical devices.

As shown in Fig. 8, the phase responses of the HPF and APF slightly deviate from the theoretical predictions at higher pole frequencies. This behavior can be mitigated by employing larger capacitance values, resulting in improved phase accuracy.

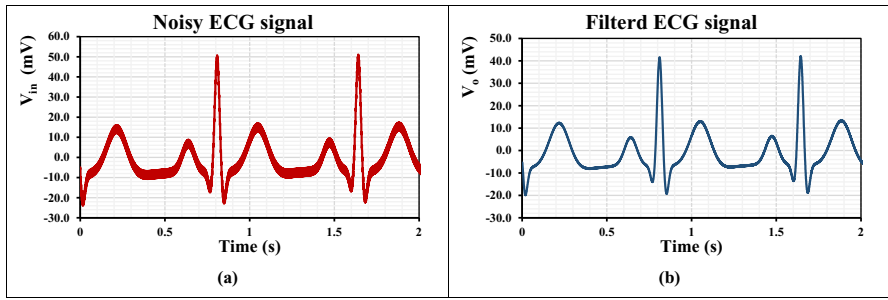


Fig. 9 Transient responses of the LPF with noisy ECG signal: **a** input and **b** output

The low-pass filter was evaluated using electrocardiogram (ECG) signals to demonstrate its potential for biomedical signal processing applications. A noisy ECG signal was created by mixing a clean ECG signal with a noise signal of 2 mV at 400 Hz, which was then applied to the input of the LP filter. The resulting noisy ECG signal is depicted in Fig. 9a. After processing through the LP filter, the output signal, showing the filtered ECG with reduced noise, is presented in Fig. 9b. The results suggest that the filter is capable of attenuating high-frequency noise while preserving the essential components of the ECG signal, indicating its potential suitability for biomedical signal processing applications.

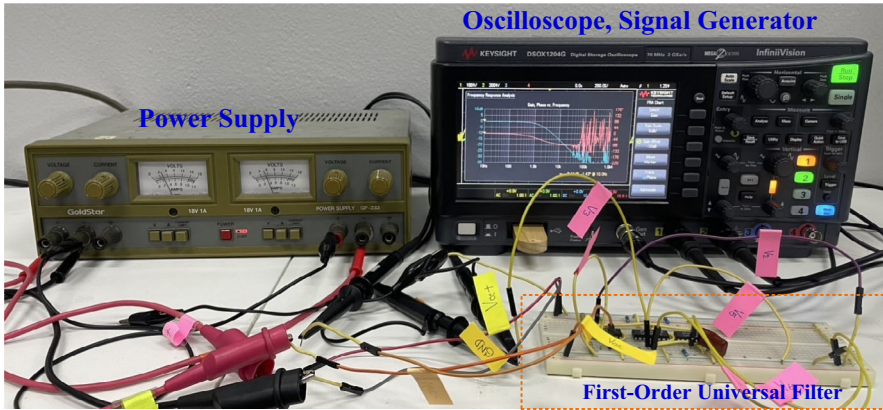
3.2 Experimental Results

The functionality of the first-order universal filter was verified through experimental testing. The prototype was implemented using commercially available LM13600N integrated circuits, with the MI-OTA depicted in Fig. 1b realized using the LM13600 IC. The first-order universal filter shown in Fig. 3 was chosen for the experiments, and the corresponding experimental setup is illustrated in Fig. 10.

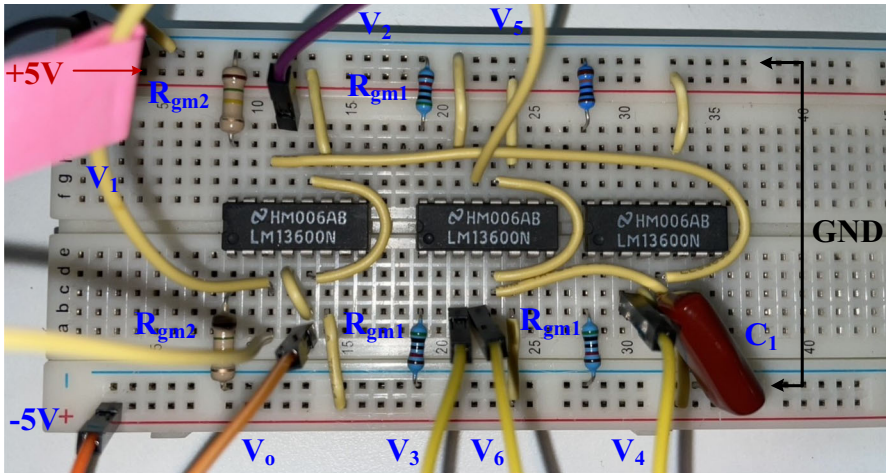
The circuit was powered with symmetric ± 5 V supply voltages, and the capacitor C_1 was set to 220 nF. A KEYSIGHT DSOX1204G oscilloscope, featuring a built-in function generator, was employed to provide the sinusoidal input signal and to observe the output waveforms. The transconductance g_{m2} was fixed at 0.496 mS using $R_{bias} = 150$ k Ω , while the transconductance g_{m1} was adjusted to control the filter's pole frequency.

In the first experiment, the transconductance was adjusted to $g_{m1} = 1.47$ mS using $R_{gm1} = 50$ k Ω . Figure 11 shows the measured magnitude and phase responses of the LPF, HPF, non-inverting APF, and inverting APF, all tuned to the same pole frequency of $f_o = 1.06$ kHz. These results verify that both non-inverting and inverting transfer functions of the APF can be realized, thereby confirming the functionality of the filter presented in Fig. 3.

To demonstrate tuning capability of the filter, the transconductance g_{m1} was varied to 0.74 mS, 1.47 mS, and 3.02 mS, by adjusting R_{gm1} to 100 k Ω , 50 k Ω , and 24 k Ω , respectively. These settings correspond to pole frequency of 535.33 Hz, 1.06 kHz, and 2.18 kHz. Figure 12 illustrates the measured magnitude responses of the LPF, HPF,



(a)



(b)

Fig. 10 Practical implementation of the first-order universal filter using MI-OTAs: **a** experimental setup; **b** close-up of the first-order universal filter circuit

non-inverting BSF, and inverting APF for different transconductance value g_{m1} . The experimental results validate the functionality of the first-order universal filter.

The phase transitions in Figs. 11c and 12c result from the phase-wrapping convention of the oscilloscope. The corresponding unwrapped phase responses vary continuously from 0° to -180° , as expected for non-inverting first-order all-pass filters.

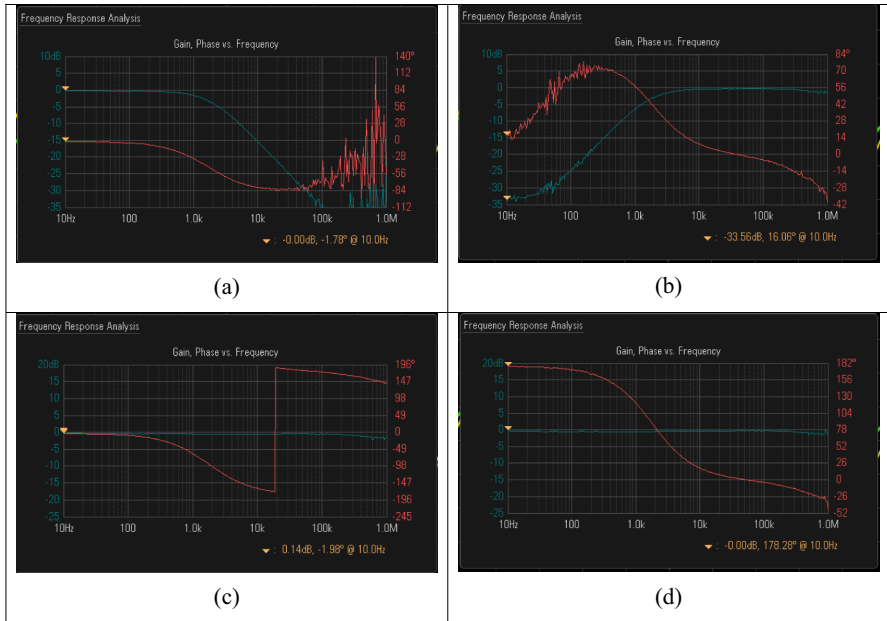


Fig. 11 Experimental frequency and phase responses of the first-order universal filter: **a** LPF, **b** HPF, **c** non-inverting APF, and **d** inverting APF, with $g_{m1}=1.47$ mS

The first-order universal filter is compared with previous works, particularly recent studies [6, 10, 12, 27, 29], and [30], as summarized in Table 3. The voltage-mode filters in [6, 10, 12, 29], which operate in a manner similar to the design, are also included. In addition, the mixed-mode filters in [27] and [30], which can also operate in voltage mode, are considered for comparison. Compared to [6, 10, 12, 27, 29], and [30], the filter offers a greater number of realizable transfer functions while operating at a lower supply voltage and reduced power consumption. The filter in [29] requires a resistance matching condition to realize the APF. The filters in [27] and [30] require capacitor-coupled input signals, making them less suitable for voltage-mode applications. Moreover, the filter in [30] requires an inverting input signal, which necessitates an additional amplifier. In contrast to [6] and [29], the filter also provides electronic tuning capability. Overall, the design demonstrates the lowest power consumption among all the filters considered. However, the filter topologies in [6] and [12] provide low output impedance, which is ideal for voltage-mode circuits, whereas the output impedance of the topology is equal to $1/g_{m2}$.

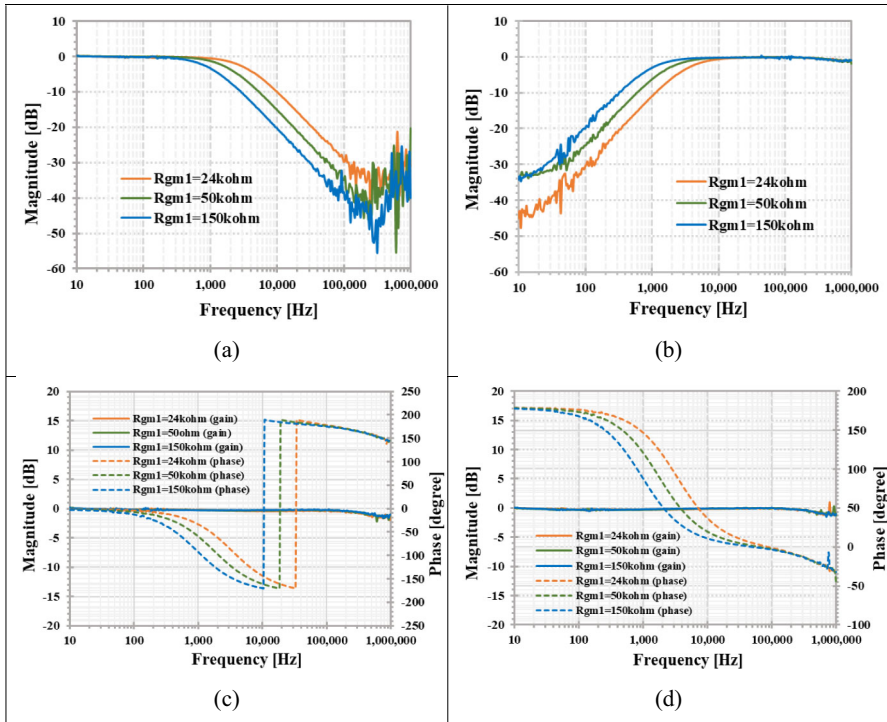


Fig. 12 Experimental frequency responses of the first-order universal filter for various g_{m1} values: **a** LPF, **b** HPF, **c** non-inverting APF, and **d** inverting APF

4 Conclusions

This paper has presented a compact, voltage-mode first-order universal filter utilizing two MI-OTAs and a single grounded capacitor. The topology enables non-inverting and inverting LPF, HPF, and APF functions, with the pole frequency being electronically tunable. By employing bulk-driven transistors in the subthreshold region, the design successfully achieves 0.5 V, 10 nW operation while avoiding the complexity of component or input matching. Both 65 nm CMOS simulations and hardware validation using commercial LM13600N ICs have confirmed the circuit's versatility and effectiveness. These results demonstrate that the MI-OTA structure is a highly efficient solution for low-power biomedical systems, offering a favorable trade-off between power consumption, design simplicity, and functional versatility.

Table 3 Comparison of the first-order universal filter with previous works

Features	Proposed	[10] 2019	[6] 2023	[27] 2023	[12] 2024	[29] 2026	[30] 2026
Active and passive elements	2-MI-OTA, 1-C	2-OTA, 1-C, 1-R	1-CFOA, 1-UGIA, 1-C, 2-R	1-VDGA, 1-C, 1-R	1-VCII, 1-C, 2-R	2-CCII, 1-C, 1-R	1-VDCC, 1-C, 2-R
Realization	CMOS structure (65 nm)	CMOS structure (180 nm)	CMOS structure (130 nm)	CMOS structure (130 nm)	CMOS structure (130 nm)	CMOS structure (180 nm)	CMOS structure (130 nm)
Mode operation	VM	VM	VM	MM	VM	VM	MM
Type of filter	MISO	MISO	MISO	MIMO	MIMO	SIMO	MISO
Number of filtering functions	6 (LPF +, LPF ₋ , HPF +, HPF ₋ , APF +, APF ₋)	3 (LPF +, HPF ₋ , APF +)	3 VM (LPF +, HPF ₋ , APF +)	12 MM 3 VM (LPF +, HPF +, APF +)	3 (LP +, LP +, AP ₋)	3 (LPF ₋ , HPF +, APF ₋)	12 MM 3 (LPF +, LPF +, APF +)
Electronic control of parameter ω_o	Yes	Yes	No	Yes	No	No	Yes
Using all grounded capacitor/resistor	Yes/Yes	Yes/Yes	No/Yes	No/Yes	No/No	Yes/No	No/Yes
Pole frequency (kHz)	0.09 (Tunable 0.045–0.34)	8.05	7960	1590	15,900	3180	987 (Tunable 987–2420)
Total harmonic distortion (%)	1 @ 140 mV _{pp}	–	~ 1 @ 266 mV _{pp}	0.15 @ 40 mV _{pp} (VM)	< 0.42	~ 0.25 @ 180 mV _{pp}	1.01 @ 60 mV _{pp}
Power supply voltages (V)	0.5	± 0.4	± 0.9	± 0.9	± 0.3	± 1.5	± 0.5
Power consumption (μW)	0.01	47.2	980	1310	–	4000	250
Verification of result	Post-layout Sim/Exp	Sim/Exp	Sim/Exp	Sim/Exp	Sim/Exp	Sim/Exp	Sim

DO-CF = Dual Output Current Follower, CCII = Second Generation Current Conveyor, VDCC = Voltage Differencing Current Conveyor (VDCC), CFOA = Current-Feedback Operational Amplifier, UGIA = Unity Gain-Inverting Amplifier, MISO = Multiple-Input Single-Output, SIMO = Single-Input Multiple-Output, MM = Mixed-Mode

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Data Availability The manuscript has no associated data.

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